

WORKED KEY

Numerical and Algebraic Expressions

This practice matches the real quiz question for question. Work every problem, then check the worked key. Anything you miss is exactly what to study.

Vocabulary: Write the term from the word bank that matches each description. Not every term is used. (1 Point Each)

numerical expression	algebraic expression	variable	term	constant	coefficient
		evaluate			

1. An expression containing at least one variable.

Answer: algebraic expression

2. A letter used to represent an unknown or changing value. _____

Answer: variable

3. A term that is a number only – its value never changes. _____

Answer: constant

4. To find the value of an expression by carrying out all of its operations.

Answer: evaluate

Problems: (2 Points Each)

5. Write a numerical expression for the verbal expression. DO NOT EVALUATE.

seven times the difference of fifteen and eight

Answer: $7(15 - 8)$ – “the difference of” groups the subtraction, so it needs parentheses

6. Write an algebraic expression for the verbal expression:

nine less than the quotient of k and 3

Answer: $k/3 - 9$ – “less than” REVERSES the order; it is not $9 - k/3$

7. Evaluate: $40 - 6 + 2^4 - 7 \cdot 5$

Answer: exponent: 16; multiply: 35; then left to right: $40 - 6 = 34$, $34 + 16 = 50$, $50 - 35 = 15$

8. Evaluate: $(5^3 - 5) / (9 + 6)$

Answer: the fraction bar groups the WHOLE top and the WHOLE bottom: $(125 - 5) \div 15 = 120 \div 15 = 8$

9. Evaluate $a^2(3b - a + 2) \div c$, if $a = 4$, $b = 3$, and $c = 8$.

Answer: $4^2(9 - 4 + 2) \div 8 = 16(7) \div 8 = 112 \div 8 = 14$

10. Error Analysis – Gus evaluated $4x^2$ when $x = 3$ and wrote 144. Identify his mistake, then evaluate correctly.

Answer: He multiplied $4 \cdot 3$ first and then squared: $(4 \cdot 3)^2 = 144$. The exponent belongs to x ONLY. Correct: $4(3^2) = 4(9) = 36$.

11. Identify the terms, the coefficients, and the constant in $7x^2 + 2x - 5$

Answer: terms: $7x^2$, $2x$, -5 ; coefficients: 7 and 2; constant: -5

12. Maya works at a café. She earns \$12 per hour making drinks and \$9 per hour at the register. She spends \$5 on lunch each day she works.

- a. Write an algebraic expression for Maya's profit if she works d hours making drinks and r hours at the register.

Answer: $12d + 9r - 5$

- b. Evaluate your expression to find her profit on a day she makes drinks for 4 hours and works the register for 3 hours.

Answer: $12(4) + 9(3) - 5 = 48 + 27 - 5 = \70